

# Leveraging LDP for High-Trust OpenSearch UBI

Keeping relevance work alive when you cannot store the queries

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# **About me**

- Independent consultant, Mountain Fog
- OpenSearch UBI and opensearch-migrations maintainer
- Apache Software Foundation member, OpenNLP PMC Chair
- PII redaction tooling at Philterd
- Works in search, NLP, and privacy

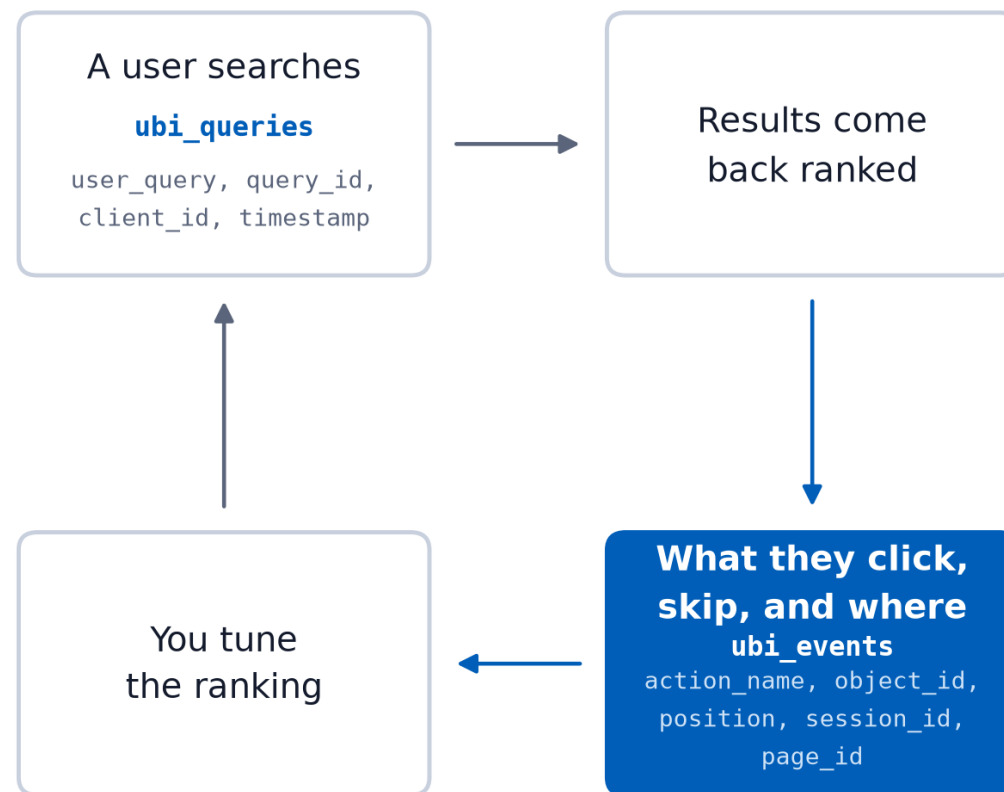
[jeffzemerick.dev](https://jeffzemerick.dev)

PART 1

# The blocker

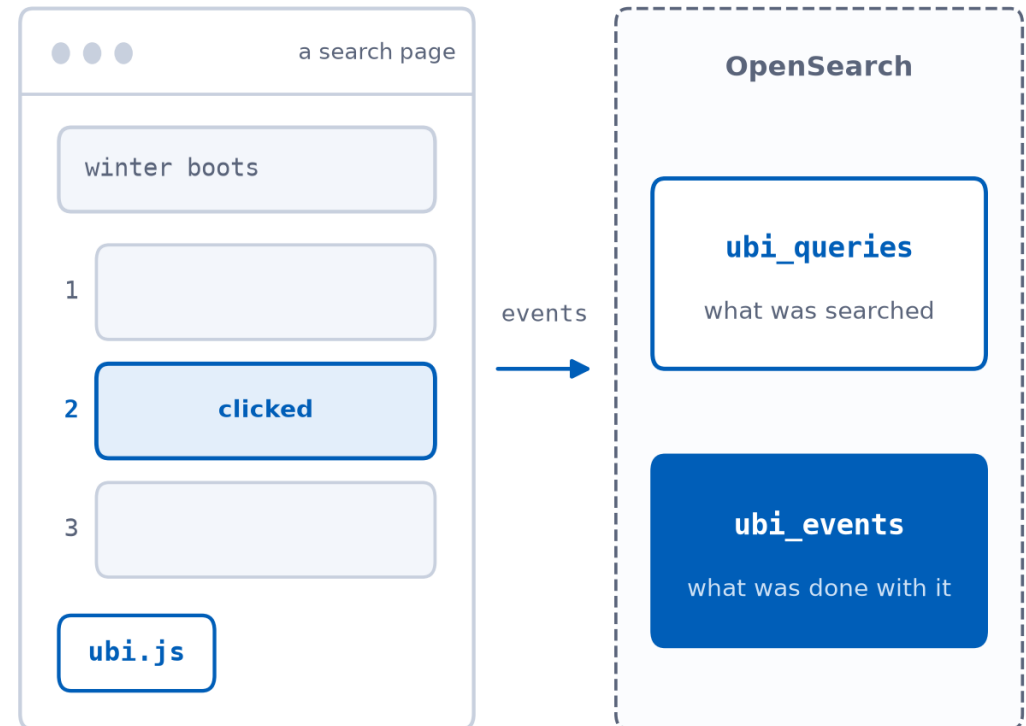
# To tune search, you have to see what users do

- Relevance work is empirical, not theoretical
- Which queries fire, which results get clicked, at which position
- Learning-to-rank trains on exactly this signal
- Without it you are guessing, and shipping guesses to production



# OpenSearch UBI exists for this

- `ubi_queries`: `user_query`, `query_id`, `client_id`, `timestamp`
- `ubi_events`: `action_name`, `object_id`, `position`, `session_id`, `page_id`
- `ubi.js` collects interactions in the browser and ships them to the cluster
- This is the raw material relevance tuning runs on



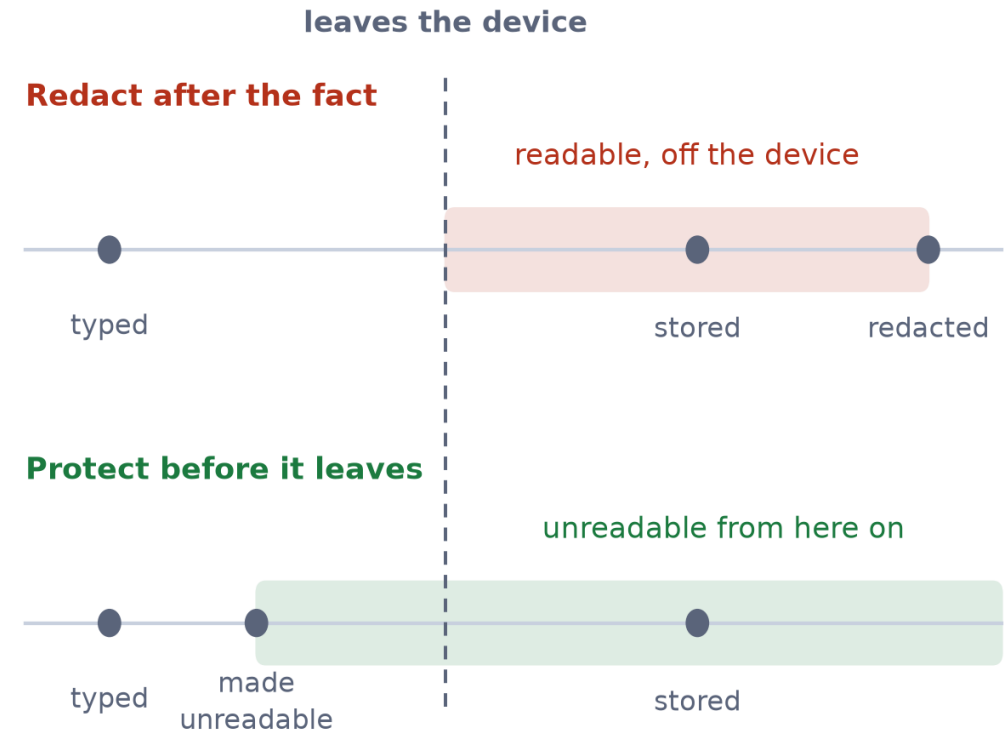
For examples and code, go to [ubisearch.dev](https://ubisearch.dev)

# Then legal reads the schema

- `user_query` is free text a person typed. It can contain anything
- `client_id` and `session_id` make it linkable across time
- In healthcare and finance, that combination is a hard stop
- Zero-trust stopped being a buzzword and became a blocker to relevance tuning

# Redaction is reactive by construction

- You collect the PII first, then try to remove it
- You discover the failure after the fact, if at all
- A free-text query has no schema to redact against
- The fix has to be that the sensitive value never leaves the device readable



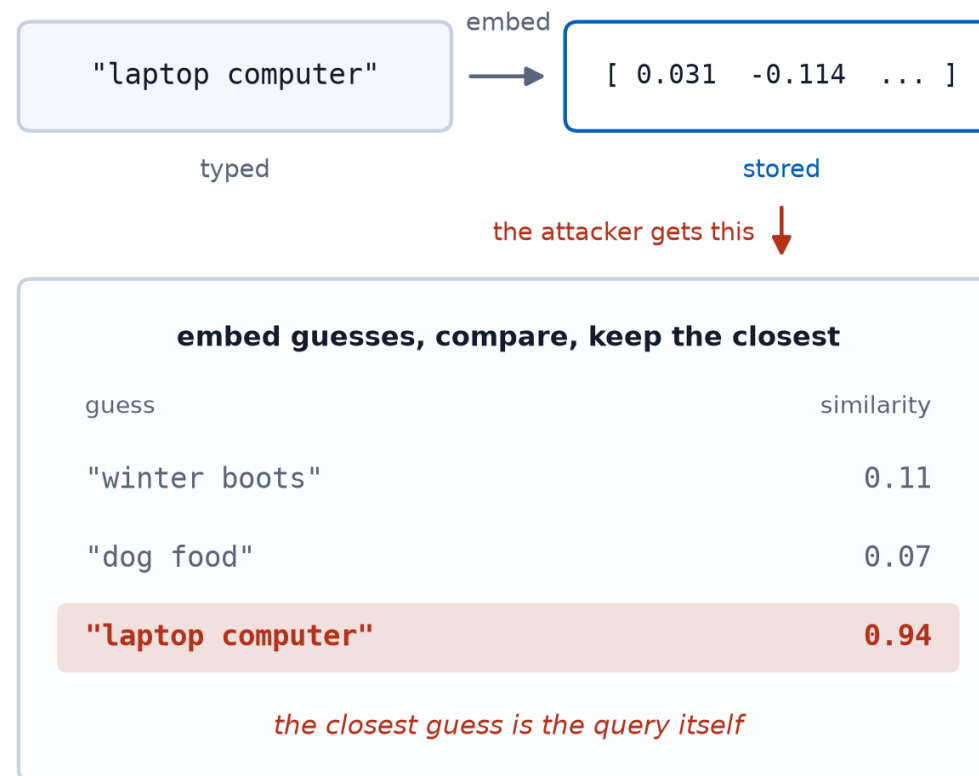
Still need to redact data at rest? [philterd.ai](https://philterd.ai)

## PART 2

# Embeddings are not anonymization

# The obvious first idea

- "We will not store the text. We will store the embedding."
- It feels private. It is a vector of floats, not words
- It is not private. It is a lossy encoding
- Vector inversion recovers the intent



LIVE NOTEBOOK

# Demo: invert a raw query vector

Notebook sections 1 to 6, then section 9

# The raw vector gives up the intent

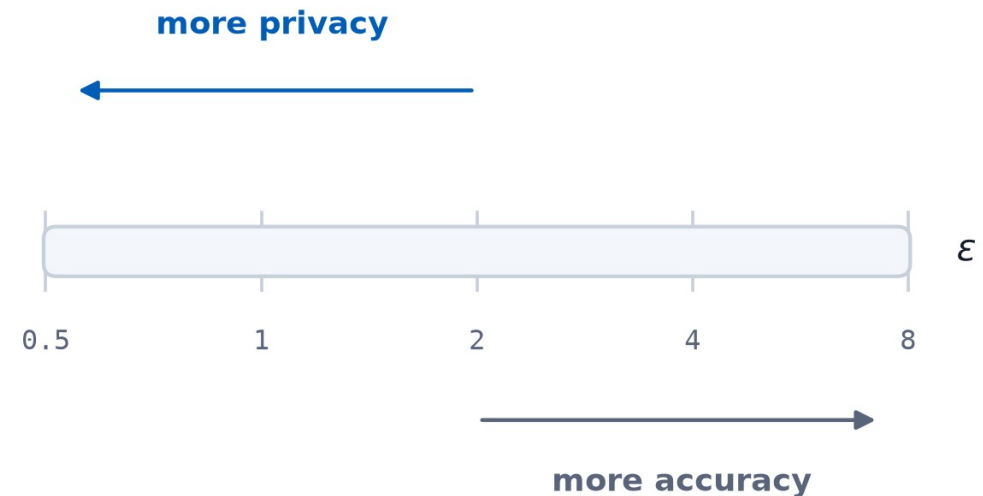
- Query: "laptop computer"
- Attacker inverts the raw vector, top hit: Laptop
- No text was stored, and the intent leaked anyway
- Storing embeddings is not a privacy control

## PART 3

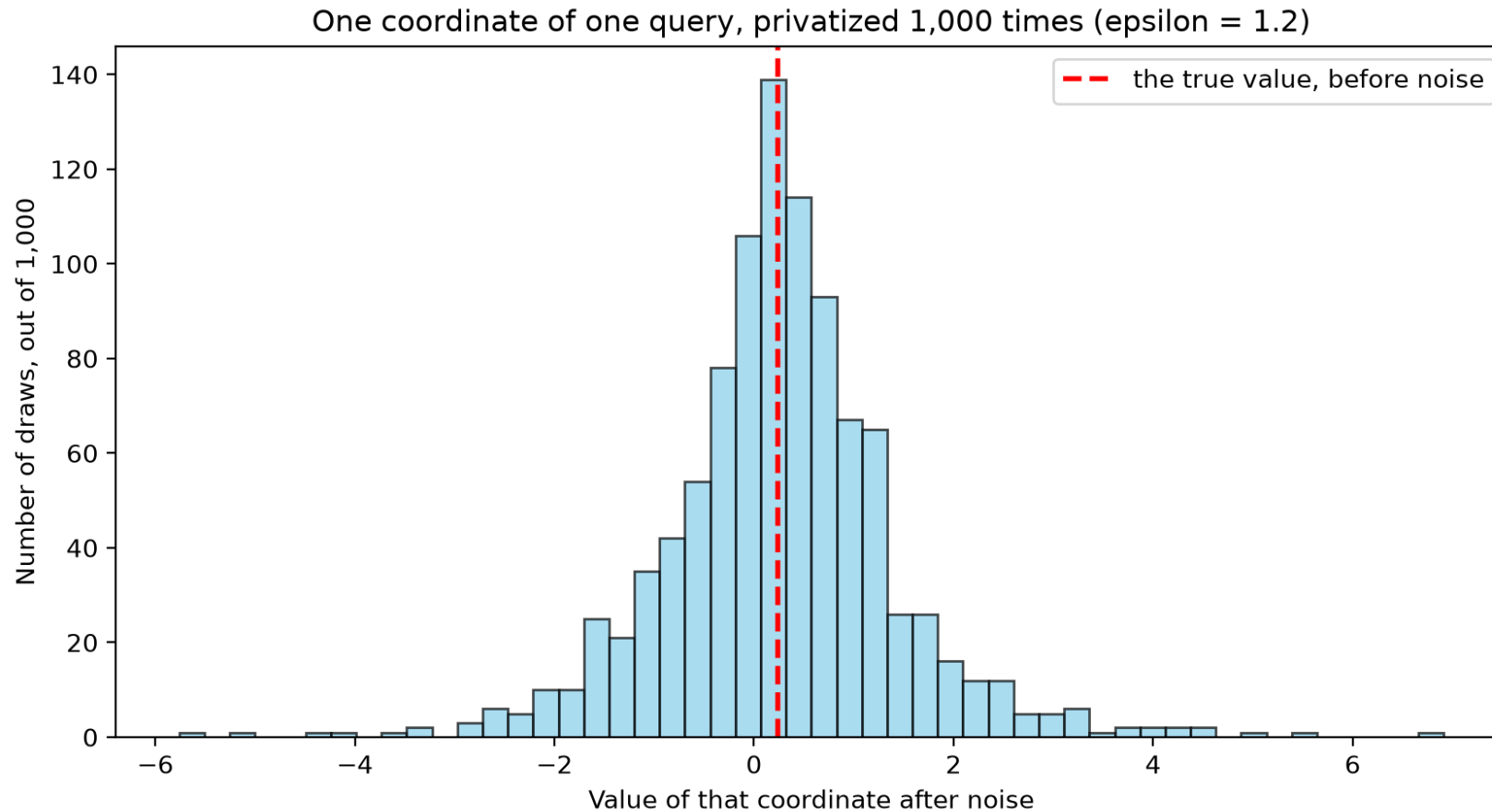
# The mechanism, and how to verify it

# Local Differential Privacy, in one slide

- Add calibrated noise to the query vector on the device
- Noise scale = sensitivity / epsilon
- Epsilon is the dial, and lower epsilon means more noise and more privacy
- The cluster never receives a readable query, so there is nothing to scrub

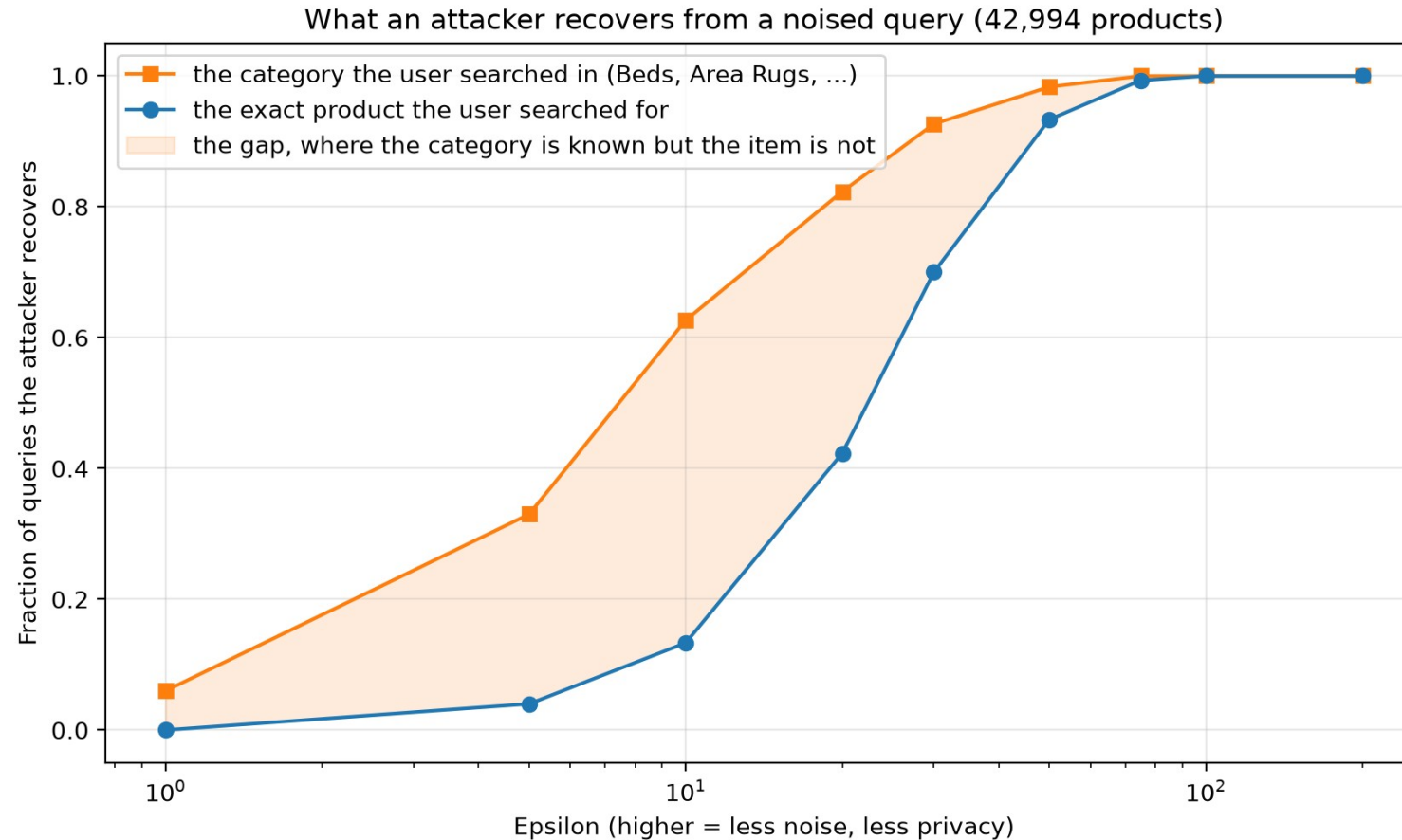


# Verification 1: the noise is what we claim



The peak sits on the true value, and the width of the tent is the privacy.

## Verification 2: measure an actual attacker



The category leaks well before the item does. At epsilon 1, neither is recoverable.

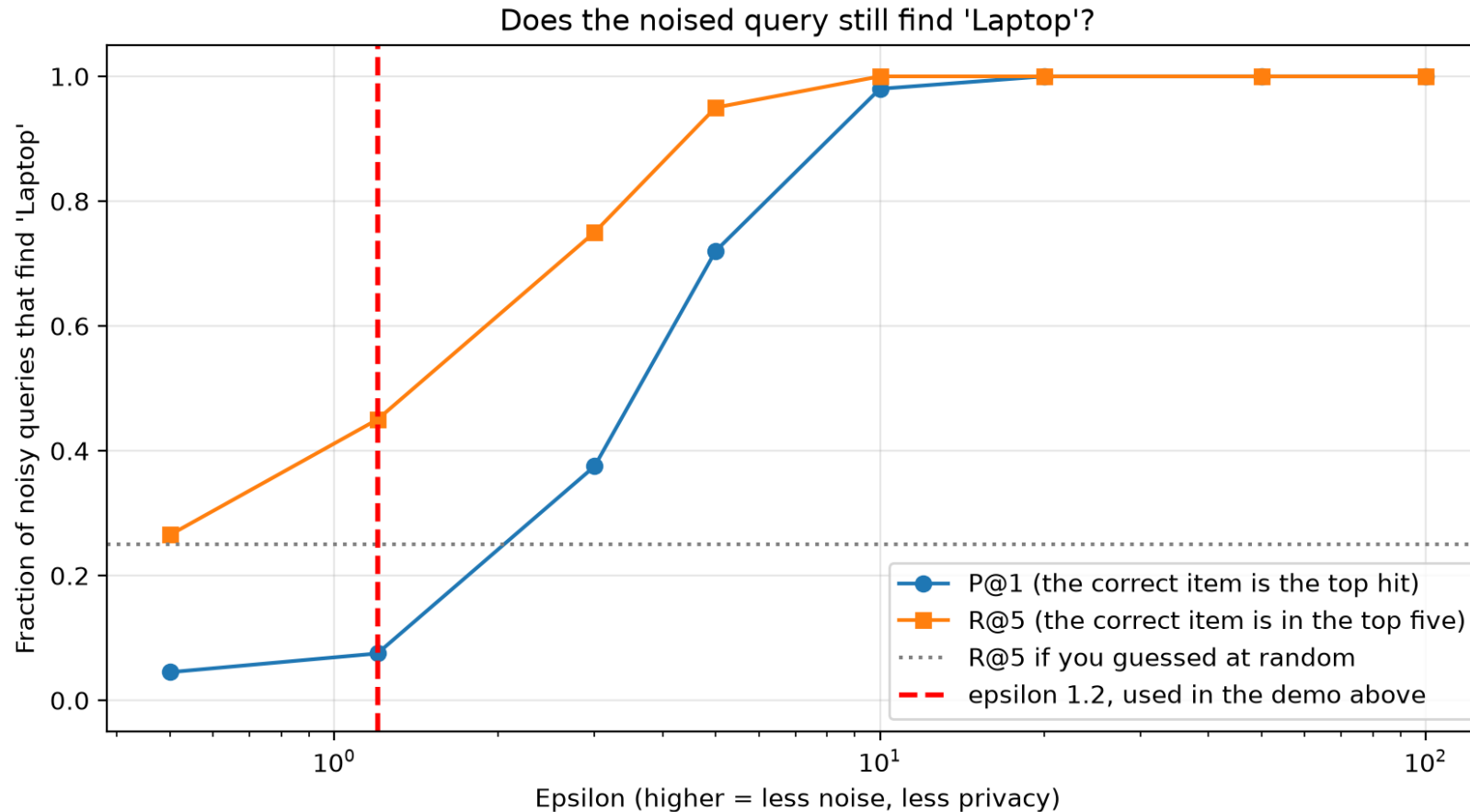
## PART 4

# What it costs

# That first demo was rigged

- It ran at epsilon 1.2 on a 20 document index
- Privacy was excellent. Utility was destroyed
- P@1 of 0.07 against a random baseline of 0.05
- The attacker learned nothing, and neither did the user

# At epsilon 1.2, barely better than guessing



The red line is where the first demo ran. Utility only comes back where the attacker wins too.

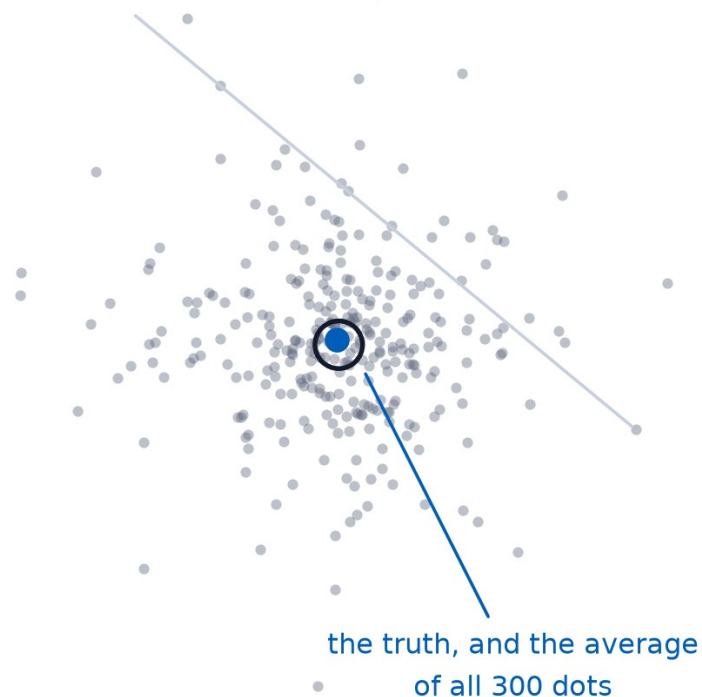
## PART 5

# What you keep

# The asymmetry is the whole point

- Laplace noise is zero-mean
- It cancels when you average over many independent users
- One person's query is unrecoverable
- The trend across ten thousand people is not

one user's noised query,  
useless on its own



LIVE NOTEBOOK

# Demo: cohorts of real users

Notebook section 11

480 real WANDS queries, grouped into 5 cohorts

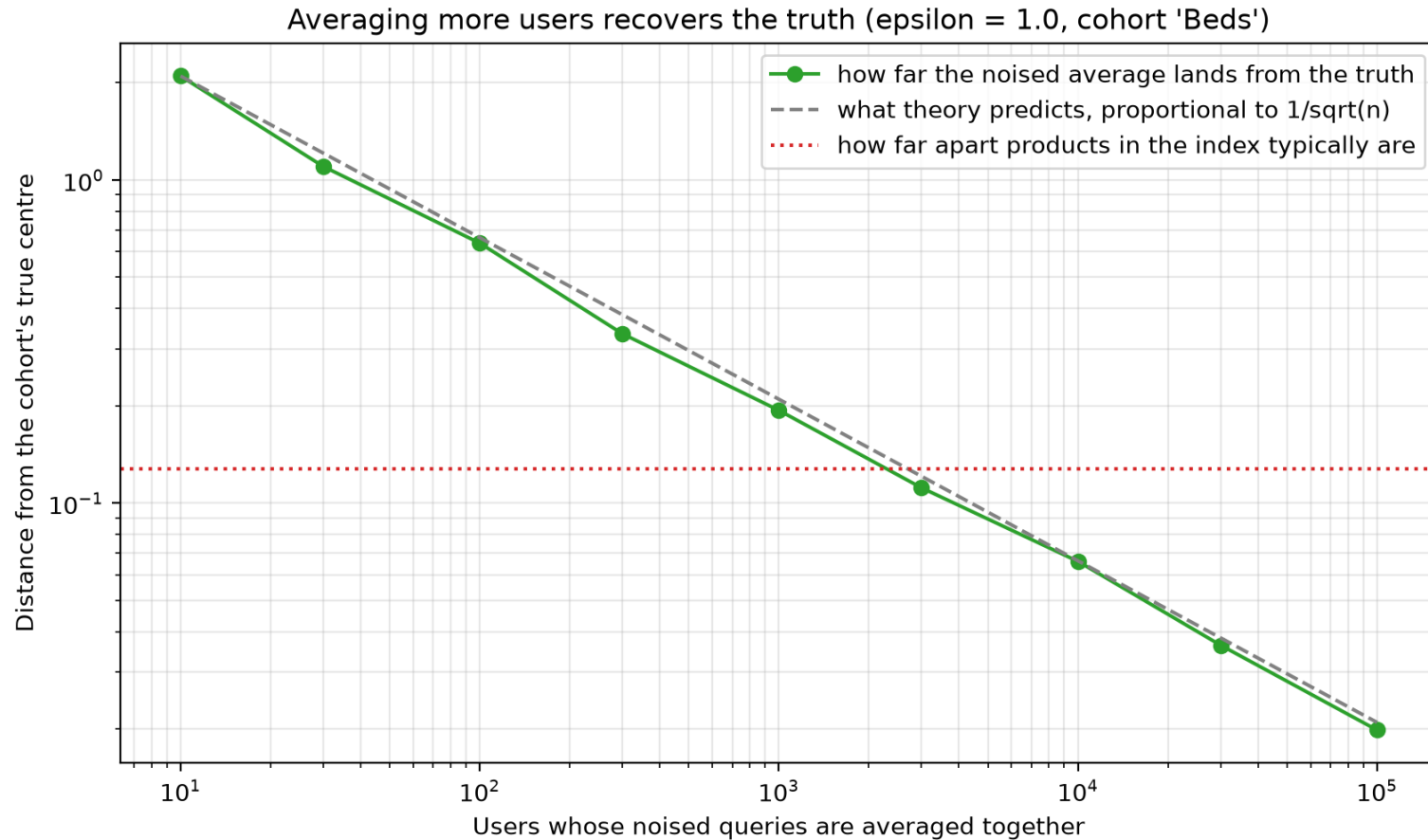
Every user privatizes independently, on their own device

# Individuals hide. The crowd does not.

| cohort                      | individual | chance | cohort ID |
|-----------------------------|------------|--------|-----------|
| Wall Art                    | 0.020      | 0.014  | OK        |
| Accent Chairs               | 0.020      | 0.027  | OK        |
| Beds                        | 0.063      | 0.026  | OK        |
| Area Rugs                   | 0.070      | 0.031  | OK        |
| Coffee & Cocktail<br>Tables | 0.030      | 0.025  | OK        |

epsilon 1.0, heavier noise than anything in Part 4. 20,000 users per cohort. 5 of 5 recovered.

# Error falls as $1 / \sqrt{n}$ , exactly as theory says



Doubling the privacy budget is expensive. Four times the users is free.

# How big a segment has to be

- Error drops below the spread of the index at roughly 2,500 users
- That is at epsilon 1, and halving epsilon needs four times the users
- Segments with thousands of users are measurable under LDP
- Segments with dozens are not measurable
- Your head terms work, but your long tail does not

**LDP does not cost you your analytics.**

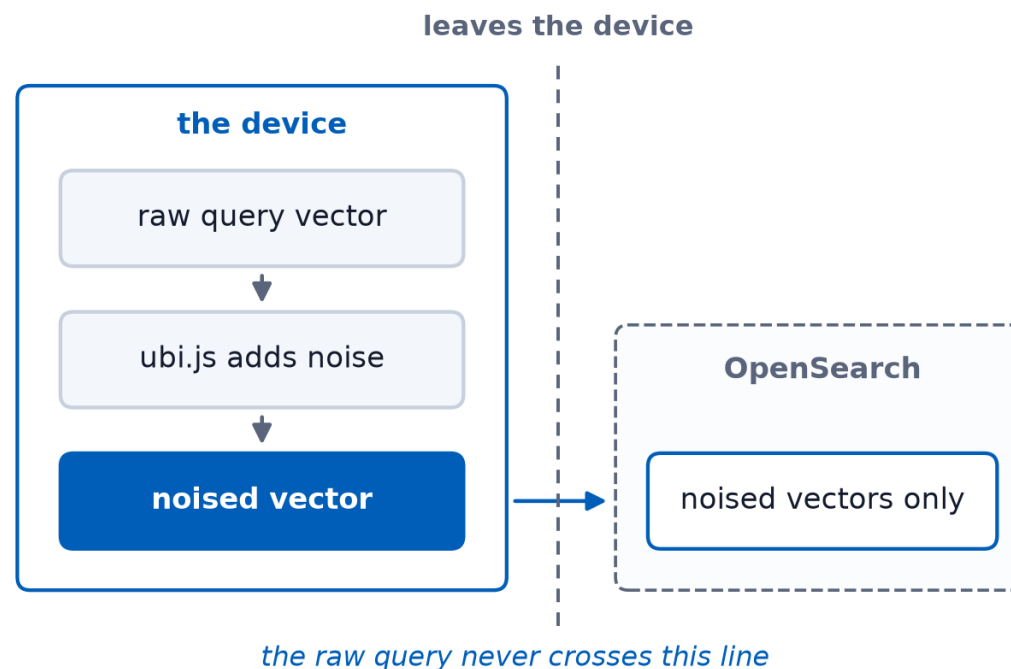
It costs you the ability to ask about one person.

## PART 6

# Back to OpenSearch, and the limits

# Where the noise gets injected

- In `ubi.js`, on the device, before the event is sent
- The cluster stores noised vectors and never sees the raw query
- Aggregation happens at query time over the noised data
- Nothing downstream needs to be trusted, because nothing downstream has the original
- You set epsilon in `ubi.js`, so anyone can read the value you shipped



# What your dashboards can still compute

- Aggregate query intent per segment
- Demand trends and shifts over time
- Cohort-level training signal for learning-to-rank
- What they cannot do is show you one user's session

# The limitations

- The category leaks. Harmless for furniture, maybe not for a medical corpus
- Epsilon does not transfer between indexes. It depends on coordinate spread
- Low-traffic segments stay unmeasurable
- Clicks are counts, not vectors, and want randomized response instead

# Takeaways

- Storing embeddings instead of text is not privacy
- Noise at the source beats scrubbing after the fact
- Verify the mechanism by attacking it, not by trusting it
- You trade individual resolution for population accuracy, and relevance only needs the latter

# Questions

[github.com/jzonthemtn/ldp-for-search](https://github.com/jzonthemtn/ldp-for-search)

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